

Introduction to Statistical Signal Processing (EC5.206) – Project

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Due date – 6th May 2026

Problem 1

You are given two ECG signals corrupted with noise (Signal1a & Signal1b). You can assume that the original ECG signals (which are not directly available) are noise-free and stationary. For each of these signals, perform noise cancellation without any reference. The method is discussed in section 9.2.5 of the textbook.

- Apply the LMS adaptive filter to perform noise removal. The parameter k_0 is not given. Qualitatively investigate the effect of the parameter k_0 ; set $n_0 = k_0 + 1$.
- Qualitatively investigate the performance as the filter order is changed from $p = 2$ to $p = 10$.
- Which of the two signals is easier to denoise? Comment on possible reasons for this.

Problem 2

We wish to approximate an unknown system with input $x[n]$ and output $y[n]$ using an LTI system of order $p = 5$. The signals are available in the file Signal2.

- Assuming jointly WSS and ergodicity, implement the SDAF filter for these inputs.
- Implement the RLS adaptive filter with an appropriate choice of forgetting factor λ . Based on your analysis, comment on the optimum value of the parameter λ for this task.
- Plot filter coefficients as a function of time for the selected λ .
- Compute and plot system approximation accuracy as function of time for $p = 5$. This accuracy can be measured using difference between the given output $y[n]$ and the LTI system output for the same input $x[n]$.
- Compute and plot overall approximation accuracy as function of filter length $p = 1$ to 10.

Problem 3

An audio signal is given to you (file Signal3). It is known that an auto-regressive model is a good approximation for this signal. Divide the signal into $M = 5$ equal portions. For each portion, use the first $N = 10\%$ to learn the AR model parameters using an LMS filter and apply it to compress the rest of the 90% of the signal. Define compression gain to be the ratio of signal norm and error norm.

- Compute and plot the (average) compression gain as a function of filter length. How would you use this to identify optimal filter length.
 - Explain using simulations – if and how any changes (increase or decrease) in M and N will affect the performance.
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